



ASSIGNMENT - 08

Deep Learning Advanced

DATA SCIENCE

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DEEP LEARNING - CNN'S

1. What is a Convolutional Neural Network (CNN) and what are CNNs primarily used for?
2. How does a convolutional layer work in a CNN?
3. What is a convolutional filter (kernel) in CNNs?
4. What are stride and padding in the context of CNNs?
5. What does a pooling layer do in a CNN and why is it used?
6. What's the difference between "same" padding and "valid" padding in CNNs?
7. What is a feature map in a CNN?
8. How does a convolutional layer differ from a fully connected layer?
9. What is transfer learning in the context of CNNs?
10. What is data augmentation and why is it important when training CNNs?
11. What are residual connections (as introduced in ResNet) and why are they helpful for deep CNNs?
12. What is the purpose of the flattening step in a CNN architecture?
13. What role does the fully connected layer play in a CNN?
14. What is the receptive field in a CNN?
15. Besides image classification, name two other tasks that CNNs are used for in computer vision.
16. What is the difference between image classification and object detection?
17. What is the difference between max pooling and average pooling?
18. CNNs use far fewer parameters than a fully connected network would on the same image data. Why is that the case?
19. If you have a 32×32 image and apply a 5×5 filter with no padding and stride 1, what will be the size of the output feature map?
20. If you apply "same" padding with a 3×3 filter on a 28×28 image with stride 1, what will the output size be?
21. How does using pooling make a CNN more robust to small translations or shifts in the input image?
22. Why do deeper CNNs (with more layers) often perform better than shallow ones for complex image tasks?
23. What are some common pre-processing steps for images before feeding them into a CNN?
24. How does the number of filters (feature maps) in a convolutional layer affect the model's capacity?
25. What are the hyperparameters of a pooling layer in a CNN?
26. Explain the two main stages of a CNN: feature extraction and classification.
27. What are "parameter sharing" and "sparse connectivity" in CNNs, and why are they beneficial?
28. Which layer in a CNN is primarily responsible for reducing the spatial dimensions of the data?
29. Can CNNs be used on data that isn't image data? Give an example.
30. What is one limitation or challenge of CNNs?